

Catching the drone you can't see

Most drone detectors listen for a radio signal. A drone flying a pre-programmed route doesn't send one. This project asks a different question — not *what is the drone saying*, but *what can its propellers not hide*?

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A plain-language companion to the technical paper.

PILLAR ONE · THE PROBLEM

The drones nobody is watching

In 2018, drone sightings shut down Gatwick Airport. Around a thousand flights were diverted and 140,000 passengers were stranded. Prisons, power stations and stadiums face smaller versions of the same problem constantly.

The standard defence is to scan for the radio chatter between a drone and its pilot. That works — right up until the drone doesn't have a pilot. A modern consumer drone can be handed a route in advance and fly it with its radio switched off entirely. To a radio scanner, it simply isn't there.

The alternatives each have their own blind spot. Radar struggles to tell a small drone from a bird. Ordinary cameras take about thirty snapshots a second, which turns a spinning propeller into a smear, and they go blind at dusk or in fog. Microphones alone are jumpy — roughly one alarm in five is a lorry, an air conditioner or a gust of wind.

There's a legal squeeze too. In most countries only a handful of agencies may lawfully jam or shoot down a drone. Everyone else — airports, prisons, security teams — is allowed to do exactly one thing: **detect**. That makes reliable, passive detection the whole game.

PILLAR TWO · THE IDEA

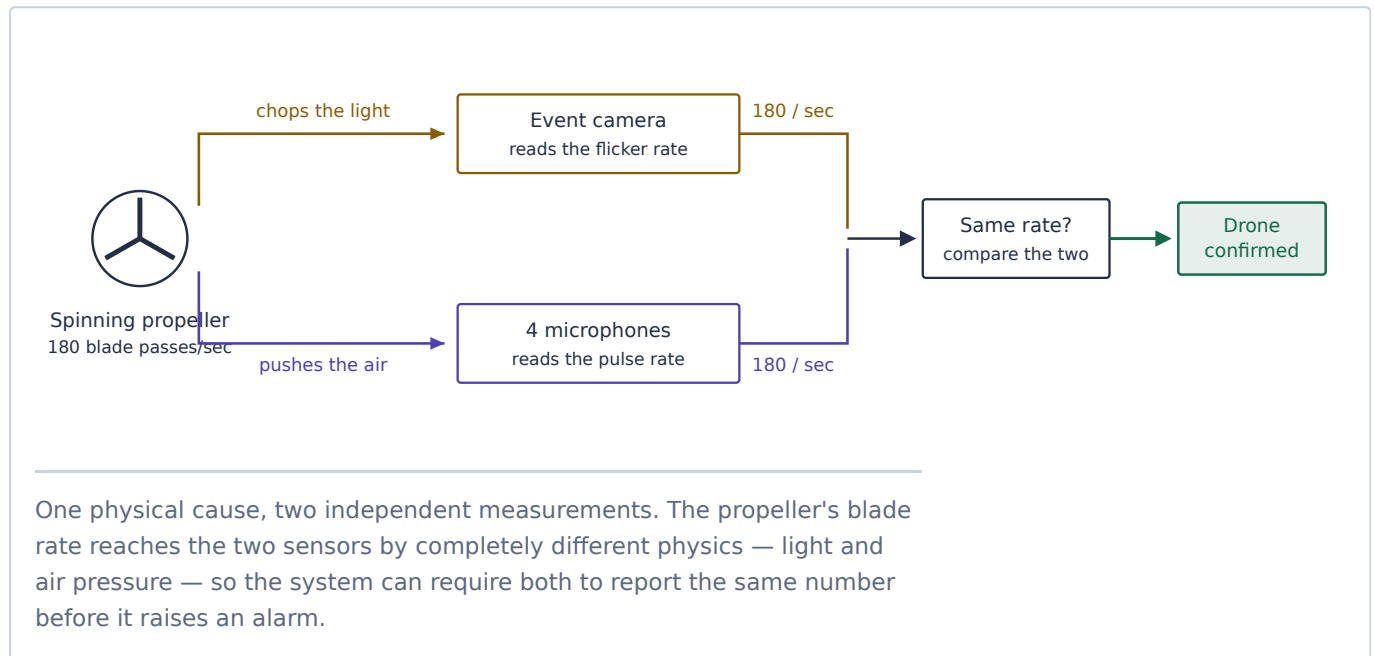
A propeller cannot hide

A drone can switch off its radio. It can be painted matte black. It cannot stop spinning its propellers — that is the only thing keeping it in the air.

And a spinning propeller does two things at once, every single time a blade goes past. It **chops the light** behind it, creating a flicker. And it **pushes the air**, creating a pulse of sound. Both happen at exactly the same rate — a few hundred times per second for a typical drone.

Engineers call that rate the *blade-pass frequency*. The useful part is that it is one physical event showing up in two completely separate places: something you could see, and something you could hear. Think of an idling car — you can watch the bonnet tremble and hear the engine hum, and both are the same pistons doing the same work.

So the system doesn't look for a drone's shape, or its colour, or its radio. It looks for that rhythm, in both places at once.



PILLAR THREE · THE INSTRUMENTS

Two senses, one of them unusual

The microphone side is ordinary enough. The camera is not — and it's the reason the idea works at all.

An ordinary camera is a flipbook: it captures a full picture thirty times a second, whether or not anything changed. A propeller turning hundreds of times a second is far too fast for that; it arrives as a blur.

An **event camera** works the way an eye does. It has no shutter and takes no photographs. Each pixel sits watching its own tiny patch of the world and calls out the instant its brightness changes — independently of every other pixel, millions of times a second. Nothing moving means nothing reported, so a camera pointed at empty sky stays almost silent until something happens in it.

THE EYE · EVENT CAMERA

Sees rhythm, not pictures

Because every pixel reports on its own schedule, the camera can follow a flicker hundreds of times per second and never blurs it. It also copes with light levels from near-darkness to direct sun — a range an ordinary camera cannot survive.

Reaction time	~1 millionth s
Light range	moonlight → sun
Motion blur	none

THE EAR · MICROPHONE RING

Hears pitch and direction

Four small microphones in a ring the size of a coaster. Comparing when a sound reaches one microphone versus another gives a rough bearing, and the propeller's steady pitch stands out from broadband background noise.

Microphones	4, in a ring
Gives bearing	yes, ±10°
Gives height	no

PILLAR FOUR · THE CROSS-CHECK

Why the false alarms die

This is the heart of it. Plenty of ordinary things flicker. Plenty of ordinary things hum. Almost nothing does both, in step, at the same rate.

A failing fluorescent tube flickers beautifully — and makes no sound at that rate. A generator hums — and doesn't flicker. Wind roars — with no steady pitch at all. Each of those can fool a single sensor, which is exactly why microphone-only systems cry wolf so often.

Requiring the two channels to *agree* throws all of them out at once. The system also allows for one sensor reporting double or triple the other's rate, which happens naturally depending on how many blades a propeller has.

	SEEN (flicker)	HEARD (pitch)	AGREE?
Drone propeller	180 / sec	180 / sec	match → DRONE
Flickering lamp	100 / sec	silent	no match → ignored
Generator / traffic	no flicker	160 / sec	no match → ignored

The everyday things that fool a single-sensor system appear in only one column. Demanding evidence in both is what turns a jumpy detector into a quiet one.

PILLAR FIVE · THE LESSON

The right design beat endless tuning

The most useful finding of the project wasn't about drones at all. It was about how the first version failed — and why no amount of adjustment could save it.

1 Split the image into a grid **FAILED**

The first design carved the camera's view into a chequerboard and looked for a rhythm inside each square. It worked — but only with the propeller a few centimetres from the lens. Further away, the handful of pixels showing the propeller were swamped by the hundreds of empty pixels sharing their square.

2 Turn up the sensitivity **FAILED WORSE**

The obvious response was to loosen every threshold. The result was 25 to 43 phantom "propellers" in every single frame. Lowering the bar didn't reveal faint drones; it just promoted background noise into confident false alarms.

3 Rebuild the approach **WORKED IMMEDIATELY**

Instead of a fixed grid, measure the rhythm of *every pixel*, then group neighbouring pixels that agree with each other. The grouping follows the propeller's actual shape, so nothing gets diluted. The very first run locked onto a single steady propeller and held it for the whole recording.

Tuning cannot repair a design that is wrong in the first place.

THE PROJECT'S CENTRAL ENGINEERING
FINDING

A second experiment sharpened the point. Allowing a *single* flickering pixel to count as evidence produced 13 to 17 false detections per frame. Requiring just **two neighbouring pixels** to agree brought that to zero. One pixel is indistinguishable from sensor noise; two that agree with each other are a real object. That single rule does most of the work of keeping the system quiet.

PILLAR SIX · WHAT EXISTS

A working prototype,
and what it measures

~100 ms to first detection — about a third of a blink	5,400 propeller RPM, read correctly off the flicker
0 false alarms across the lab recordings	0.63 ms to analyse one frame after the rewrite

The system runs live, watches a propeller, reports its speed in revolutions per minute, and raises a confirmed detection in about a tenth of a second. It grows more confident the longer it keeps seeing the same thing, and quietly forgets what stops appearing — so a brief glitch fades instead of latching on as an alarm.

It was first built in Python to get the ideas right, then rewritten in C++ so that each frame takes a predictable amount of time rather than occasionally stalling. For something meant to run unattended for months, steady and boring is the goal.

PILLAR SEVEN · THE HONEST PART

What this does not yet prove

Everything above was measured indoors, on a bench fan standing in for a drone, in recordings around thirty seconds long. That matters, and the project documents its failures as carefully as its successes.

- **No outdoor test, and no real drone yet.** A workshop is not a windy car park at dusk. Until it has faced one, the results describe a promising method, not a finished product.
- **The published range figures are estimates.** Distances of 50 to 150 metres are calculated from the physics and the sensor's resolution — they have not been measured in the field, and the paper says so plainly.
- **Matching *direction* between the two sensors did not work.** Matching the rhythm works reliably. Getting the camera and the microphones to agree on *where* the source was did not: in an indoor room the microphones barely registered any swing while the camera swept across 25

degrees. Reflections off the walls are the likely culprit. It is reported as a genuine negative result rather than quietly dropped.

- **Only one target at a time.** The system currently assumes a single source. Several drones at once, at similar speeds, would need more work.

None of this undermines the core finding. The physics holds, the cross-check works, and the detector is fast and quiet. What comes next is taking it outside and pointing it at something that is actually flying.

Companion to “Passive RF-Silent Drone Detection via Neuromorphic Vision and Acoustic Fusion of the Propeller Blade-Pass Frequency.”

Every figure here is drawn from measurements recorded in the project's own logs.